

HILLSDALE, MI, USA

WMO: 725409

Lat: 41.921N

Lon: 84.586W

Elev: 1182

StdP: 14.08

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.3	5.3	-7.5	3.8	2.4	-1.8	5.2	7.4	23.4	24.1	20.6	24.9	5.1	240	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	89.0	72.7	86.1	71.4	83.5	70.3	76.6	84.1	74.9	82.0	73.3	79.9	8.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	133.8	80.6	72.6	126.2	78.9	71.6	121.9	77.9	40.9	84.5	39.2	81.8	37.6	80.3	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.3	18.1	16.2	-6.4	91.9	5.8	3.0	-10.6	94.1	-14.0	95.8	-17.2	97.5	-21.5	99.7		
			-7.0	79.0	5.7	2.1	-11.1	80.5	-14.5	81.7	-17.7	82.9	-21.8	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	24.0	26.1	36.1	48.3	59.4	68.1	71.6	69.5	63.6	51.6	40.0	29.5
	DBStd	19.04	11.17	10.77	11.16	9.34	8.58	6.53	5.88	5.21	7.47	8.99	9.56	10.04
	HDD50	3141	808	672	452	139	16	0	0	0	2	91	324	637
	HDD65	6440	1272	1090	896	506	220	41	10	18	113	424	750	1100
	CDD50	2816	1	2	22	88	306	544	669	604	412	142	24	2
	CDD65	641	0	0	2	5	45	135	214	157	73	10	0	0
	CDH74	5350	0	0	13	47	406	1117	1904	1178	609	76	0	0
	CDH80	1489	0	0	2	5	89	329	616	271	166	11	0	0
Wind	WSAvg	6.8	8.2	8.2	8.2	8.8	7.0	5.7	4.7	4.2	4.9	6.5	7.6	8.1
Precipitation	PrecAvg	35.2	2.0	1.7	2.7	3.3	3.6	3.8	3.6	3.3	3.2	2.6	2.8	2.5
	PrecMax	43.5	5.0	4.5	5.4	5.5	7.2	8.2	8.1	7.3	6.4	8.6	6.9	5.1
	PrecMin	22.9	0.2	0.3	0.7	1.0	1.4	0.5	1.0	0.6	0.3	0.2	0.8	0.7
	PrecStd	4.7	1.3	1.0	1.1	1.1	1.4	1.6	1.7	1.8	1.6	1.6	1.3	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.9	58.6	74.6	79.4	87.0	90.5	93.6	89.8	89.6	81.6	70.0	58.6
		MCWB	52.1	52.6	62.8	61.2	69.5	71.6	74.5	73.7	71.3	67.9	57.8	56.8
	2%	DB	48.5	52.3	64.4	73.4	82.1	87.7	89.8	85.8	84.5	75.1	63.3	52.8
		MCWB	46.3	49.3	55.8	59.1	66.0	71.8	74.1	72.3	69.9	62.6	54.8	49.7
	5%	DB	44.5	45.5	58.8	69.8	79.1	83.8	86.3	82.4	81.1	70.6	59.0	46.8
		MCWB	41.2	42.0	51.9	56.2	65.1	69.3	72.7	71.0	67.9	61.1	52.6	44.0
	10%	DB	37.9	39.5	53.7	64.1	74.5	81.0	82.5	80.8	77.1	66.2	54.8	42.8
		MCWB	35.5	35.8	47.4	53.6	63.4	68.7	71.0	69.9	66.2	59.2	49.4	40.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	54.3	63.2	66.9	72.7	77.0	80.0	77.8	75.3	70.7	60.6	56.8
		MCDB	54.5	55.7	75.0	75.9	80.6	85.0	86.6	85.6	83.6	78.8	66.6	58.3
	2%	WB	46.5	48.6	57.7	62.6	70.1	74.8	77.5	75.4	73.2	66.5	57.0	50.4
		MCDB	47.5	52.0	63.3	69.9	77.5	82.4	85.1	81.9	80.7	71.7	61.5	52.1
	5%	WB	40.9	42.3	53.4	58.7	67.8	72.9	75.4	73.9	71.1	63.4	53.6	44.7
		MCDB	43.9	46.3	58.4	65.8	75.8	79.9	82.8	79.7	77.0	67.4	57.7	46.5
	10%	WB	36.3	36.7	47.2	55.1	65.4	70.9	73.5	71.9	68.6	59.9	49.8	39.9
		MCDB	38.5	39.8	52.3	63.4	72.3	77.6	80.2	77.7	74.3	65.7	55.1	42.5
Mean Daily Temperature Range	5% DB	MDBR	12.9	13.9	16.2	19.3	19.4	18.8	19.7	19.1	20.5	18.0	14.3	11.4
		MCDBR	16.3	18.2	23.2	25.1	23.4	22.0	21.5	21.1	23.9	23.4	19.5	15.8
		MCWBR	14.3	15.1	16.0	14.2	12.1	9.9	9.6	10.9	11.2	13.0	13.8	13.9
	5% WB	MCDBR	15.5	18.1	21.3	21.9	20.2	18.3	18.2	17.2	19.0	19.4	18.2	15.1
		MCWBR	14.2	15.9	16.3	15.4	12.4	10.6	10.5	10.3	10.9	13.7	14.9	14.4
Clear-Sky Solar Irradiance	taub	0.299	0.342	0.347	0.388	0.416	0.435	0.425	0.424	0.380	0.345	0.314	0.300	
	taud	2.378	2.220	2.345	2.253	2.217	2.202	2.241	2.238	2.345	2.437	2.475	2.408	
	Ebn at Noon	268	270	285	279	273	267	269	265	270	268	262	257	
	Edh at Noon	26	35	35	42	44	45	43	42	35	28	23	23	
All-Sky Solar Radiation	RadAvg	480	767	1124	1459	1696	1908	1934	1676	1364	890	562	394	
	RadStd	49	104	107	132	156	124	97	78	119	82	77	43	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page